

Conclusion /inference

In Lab 2, we found that air quality gets really bad during winter (November and December) because of stubble burning after harvest. When we merged the air data and crop data at the state level, it showed a weak negative correlation of -0.20 . This means states with more pollution have slightly lower rice yields because toxic air damages the crop plants.

From the optional tasks, we found a weird paradox: highly polluted states like Punjab and Haryana actually produce much higher rice yields than clean states like Mizoram. This happens because Punjab/Haryana use intensive farming, canal irrigation, and heavy fertilizers which hides the bad effect of pollution. This proves that correlation is not causation, and factors like water and soil quality are much more important for crop yields.